Research Prizes with Uncertain Compute Costs: From Entry Games to Brownian Stopping Races

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Abstract

We study competition for a research prize when the required computation is uncertain and researchers have different costs. A one-period model gives explicit mixed-entry conditions and an exact potential, hence a pure equilibrium also exists. A continuous-time extension replaces committed attempts by privately observed diffusions with irreversible abandonment; the first solution wins. Against fixed rival strategies, their solution hazard acts as additional discounting. We solve the constant-hazard, driftless case in closed form and formulate the general equilibrium as coupled stopping and forward-distribution equations. This connects research races to Brownian bandit pruning, while distinguishing a strategic fixed point from a planner's optimization problem.

1 Setting and relation to the literature

A large mathematical problem can be worth attempting alone even when many potential solvers prefer to stay out of a prize contest. The relevant cost is not necessarily a known token budget: it may be the random expenditure incurred before success, abandonment, or a rival's solution. The question is how competition changes participation and continued computation.

The one-period benchmark belongs to the literature on endogenous contest entry. Morgan, Orzen, and Sefton [2] study entry and subsequent rent-seeking, including the role of the outside option. Their post-entry effort game differs from our fixed-attempt lottery. Fu, Jiao, and Lu [1] analyze complete-information lottery contests with costly entry and subsequent bidding; we instead hold each attempt's success probability fixed and allow researcher heterogeneity. Fullerton and McAfee [3] study heterogeneous research contestants and selection through auctioned entry, rather than the uncoordinated binary entry considered here. Our finite-game potential is in the tradition of Rosenthal [4] and Monderer and Shapley [5]. These connections locate the benchmark; we make no novelty claim for entry indifference or the potential-game method.

The dynamic model belongs to the research-race tradition represented by Loury [6]. Its computational connection is to the Brownian bandit framework [7]: a backward continuation problem and a forward population calculation are coupled. Here the coupling is strategic and the payoff is first passage to a solution, rather than a terminal maximum. We prove the elementary entry and constant-hazard results below. The general dynamic equilibrium is a characterization, not an existence, uniqueness, or numerical convergence theorem.

2 One-period entry with heterogeneous costs

There are $n \geq 2$ risk-neutral researchers and a prize $V > 0$. Researcher i enters with probability q_i, incurs a random attempt cost with finite mean $k_i > 0$, and succeeds with probability $p_i \in (0, 1]$. Parameters are common knowledge. Entry randomizations and successes are independent across

researchers. Own cost and own success may be dependent. An entrant commits to the entire attempt; nonentry pays zero. If several entrants succeed, one is selected uniformly to receive V; if none succeeds, no prize is paid. Assume $k_i < Vp_i$, so each attempt is profitable alone.

Proposition 1 (Entry equilibrium)**.** *Define*

$$D_i(q_{-i}) = Vp_i \int_0^1 \prod_{j \neq i} (1 - q_j p_j + q_j p_j z)\, dz - k_i. \tag{1}$$

A profile $q \in [0,1]^n$ is a Nash equilibrium exactly when

$$q_i = 0 \Rightarrow D_i \leq 0, \qquad 0 < q_i < 1 \Rightarrow D_i = 0, \qquad q_i = 1 \Rightarrow D_i \geq 0 \tag{2}$$

for every researcher.

Proof. Let K_{-i} count successful opponents. Conditional on own success, the prize probability is $(1 + K_{-i})^{-1}$. Since $1/(k+1) = \int_0^1 z^k\, dz$, independence gives the integral in (1). Expected payoff is $q_i D_i(q_{-i})$; optimizing this linear function on $[0,1]$ yields (2). □

Two researchers. The fully mixed equilibrium, when it exists, is

$$q_1 = \frac{2}{p_1}\left(1 - \frac{k_2}{Vp_2}\right), \qquad q_2 = \frac{2}{p_2}\left(1 - \frac{k_1}{Vp_1}\right). \tag{3}$$

Both probabilities are interior exactly when $Vp_i(1 - p_j/2) < k_i < Vp_i$ for $i \neq j$. For $V = 100$, $p_1 = p_2 = 1$, and $(k_1, k_2) = (60, 80)$, the solution is $(q_1, q_2) = (0.4, 0.8)$. The cheaper researcher enters less often in this mixed equilibrium because each probability makes the *opponent* indifferent. Both $(1,0)$ and $(0,1)$ are also equilibria. The example does not imply a general negative relationship between efficiency and participation across selected equilibria.

Symmetric benchmark. For common p, k, the gross entry return against common probability $q > 0$ is

$$R(q) = \frac{V[1 - (1 - pq)^n]}{nq}, \qquad R(0) = Vp. \tag{4}$$

It is strictly decreasing. Under $0 < k < Vp$, the unique symmetric equilibrium has $q = 1$ if $k \leq V[1 - (1-p)^n]/n$; otherwise its interior probability solves $R(q) = k$. For $p = 1$, universal entry requires $k \leq V/n$. In an interior symmetric equilibrium, $nqk = V[1 - (1 - pq)^n]$: expected expenditure equals expected prize payout.

Proposition 2 (An exact potential)**.** *For a pure entrant set A, let B_i be independent Bernoulli variables with parameters p_i, and let $H_m = \sum_{\ell=1}^{m} 1/\ell$, with $H_0 = 0$. Then*

$$\Phi(A) = V\mathbb{E}\Big[H_{\sum_{i \in A} B_i}\Big] - \sum_{i \in A} k_i \tag{5}$$

is an exact potential. In particular, a pure Nash equilibrium exists.

Proof. Adding $i \notin A$ changes (5) by $Vp_i\mathbb{E}[(1 + \sum_{j \in A} B_j)^{-1}] - k_i$, exactly the entrant's payoff change. A maximizing entrant set exists because the set of profiles is finite, and admits no profitable unilateral deviation. □

Thus mixing is not necessary for existence. When $p_i = 1$, the potential reduces to $VH_{|A|} - \sum_{i \in A} k_i$, the harmonic form associated with a congested common resource. The potential is not social welfare: aggregate expected payoff is $V[1 - \prod_{i \in A}(1 - p_i)] - \sum_{i \in A} k_i$.

3 A continuous-time race with private progress

We now change the allocation rule: the *first* verified solution wins. Researcher i starts at $x_i < b$ and, while working, observes

$$dX_i(t) = \mu_i\,dt + \sigma_i\,dW_i(t), \qquad \sigma_i > 0. \tag{6}$$

The Brownian motions are independent. Hitting b produces a solution; working costs $c_i > 0$ per unit time. Each researcher can abandon irreversibly, but cannot pause, restart, or change compute intensity in this baseline. Everyone discounts at rate $r \geq 0$ and observes the first solution announcement immediately. All parameters and initial-state distributions are common knowledge; realized rival states and abandonment decisions are unobserved. Initial states and private randomizations are independent across researchers.

The state X_i is a stylized measure of an approach's promise, not a claim that proof completion has a literal observable percentage. The model requires such a state to summarize the researcher's predictive information. Common uncertainty about problem difficulty would violate independence and require additional beliefs.

For each strategy, define its *potential* solution time T_i by running it as though no rival ever announced success; set $T_i = \infty$ if it abandons first. Let θ_i be the corresponding abandonment time and $T_{-i} = \min_{j\neq i} T_j$. Actual payoff is

$$\mathbb{E}\left[Ve^{-rT_i}\mathbf{1}_{\{T_i<T_{-i},\ T_i<\infty\}} - \int_0^{\theta_i\wedge T_i\wedge T_{-i}} e^{-rt}c_i\,dt\right]. \tag{7}$$

Nondegenerate diffusions started below b have no finite-time ties in this setting. Infinite times in (7) contribute no prize. Unlike Section 2, costs cease when a rival wins.

3.1 Best response to a rival solution hazard

Fix rival strategies and let

$$h_{-i}(t) = -\frac{d}{dt}\log\mathbb{P}(T_{-i} > t)$$

where the survival probability is positive and differentiable. Independence makes this a deterministic function of time, conditional on the prize remaining available. Define $\mathcal{L}_i = \mu_i\partial_x + (\sigma_i^2/2)\partial_{xx}$ and $D_i(t,s) = \exp\{-\int_t^s [r + h_{-i}(u)]\,du\}$. Conditional on time t, own state x, and no prior solution, the value is

$$v_i(t,x) = \sup_{\theta\geq t}\mathbb{E}_{t,x}\left[VD_i(t,\tau_b)\mathbf{1}_{\{\tau_b\leq\theta,\ \tau_b<\infty\}} - \int_t^{\theta\wedge\tau_b} D_i(t,s)c_i\,ds\right], \tag{8}$$

where τ_b is own first passage to b and immediate abandonment is allowed. Success takes precedence at b.

Dynamic programming gives the obstacle equation

$$\boxed{\max\{-v_i,\ \partial_t v_i + \mathcal{L}_i v_i - (r + h_{-i})v_i - c_i\} = 0, \qquad v_i(t,b) = V.} \tag{9}$$

The value is the bounded optimal-stopping value in (8); where classical regularity fails, (9) is interpreted in the usual weak or viscosity sense. For a deadline T, add $v_i(T,x) = 0$ for $x < b$. Competition acts as additional discounting, but its rate must ultimately be consistent with rival behavior.

4 An exact constant-hazard benchmark

Take one researcher with $\mu = 0$, cost rate c, volatility σ, an infinite horizon, and constant rival hazard h. Write $\rho = r + h$.

Proposition 3 (Optimal abandonment boundary)**.** *For $\rho > 0$, define*

$$\kappa = \frac{\sqrt{2\rho}}{\sigma}, \qquad a = b - \frac{1}{\kappa}\operatorname{arcosh}\left(1 + \frac{\rho V}{c}\right). \tag{10}$$

The optimal policy abandons at $x \le a$ and continues on (a, b). Its value is

$$v(x) = \begin{cases} 0, & x \le a, \\ \dfrac{c}{\rho}\big[\cosh(\kappa(x-a)) - 1\big], & a < x < b, \\ V, & x = b. \end{cases} \tag{11}$$

For $\rho = 0$, the limiting solution has $a = b - \sigma\sqrt{V/c}$ and $v(x) = c(x-a)^2/\sigma^2$ on (a, b).

Proof. On (a, b), solve $(\sigma^2/2)v'' - \rho v - c = 0$ with $v(a) = v'(a) = 0$. This yields (11); $v(b) = V$ gives (10). The function is nonnegative, bounded by V, and continuously differentiable at a. In the stopping region the continuation operator equals $-c < 0$, while in the continuation region it equals zero. Discounted Itô verification therefore bounds every admissible stopping payoff above by v, with equality for exit from (a, b). For $\rho > 0$ the bounded terminal remainder vanishes; for $\rho = 0$ the proposed interval exit has finite expected duration and the quadratic candidate verifies directly. Finally, $\operatorname{arcosh}(1 + z) \sim \sqrt{2z}$ gives the stated limit. □

The continuation distance $b - a$ increases with V and decreases with c and ρ. For the last claim, put $z = \rho V/c$ and observe that $\operatorname{arcosh}(1+z)/\sqrt{z}$ decreases for $z > 0$. Indeed, writing $y = \operatorname{arcosh}(1+z)$ reduces its derivative sign to $2\tanh(y/2) < y$. Thus greater competition raises the abandonment threshold. These are best-response comparative statics at a prescribed hazard, not comparative statics of the full equilibrium. A constant positive hazard is a benchmark; it need not be generated by a finite population of abandoning researchers.

5 Closing the equilibrium loop

For deterministic time–state stopping policies, potential solution times remain independent. Let

$$F_i(t) = \mathbb{P}(T_i \le t), \qquad f_i(t) = F_i'(t), \qquad S_i(t) = 1 - F_i(t). \tag{12}$$

The distributions can be defective: $S_i(\infty) > 0$ when some paths never solve. Then

$$\boxed{h_{-i}(t) = \sum_{j \ne i} \frac{f_j(t)}{S_j(t)}.} \tag{13}$$

To see this, $\mathbb{P}(T_{-i} > t) = \prod_{j \ne i} S_j(t)$; logarithmic differentiation proves (13). The same construction allows independent private mixtures over policies.

Forward calculation. For a regular continuation region $(a_i(t), b)$, let $m_i(t, x)$ be the unconditional density of the potential process still working and unsolved. It satisfies

$$\partial_t m_i = -\mu_i \partial_x m_i + \frac{\sigma_i^2}{2} \partial_{xx} m_i, \qquad m_i(t, a_i(t)) = m_i(t, b) = 0, \tag{14}$$

with the specified initial law and absorption at either boundary. The solution density is the upper-boundary flux

$$f_i(t) = -\frac{\sigma_i^2}{2} \partial_x m_i(t, b^-). \tag{15}$$

For nonsmooth or jumping stopping regions, use the killed transition law and account explicitly for newly abandoned mass instead of assuming classical boundary regularity.

Two bookkeeping points matter. First, $S_i(t)$ includes abandoned paths; it is *not* $\int m_i(t, x)\, dx$. Second, (14) does not kill mass at the rival hazard: it describes potential paths without rival announcements. Including such killing here and also in (9) would count the rival's termination twice.

Equilibrium characterization. A consistent profile consists of strategies optimal for (8), together with their forward laws and hazards satisfying (12)–(13). Such a profile is a Nash equilibrium under the stated private-information structure: fixing rivals fixes their hazard, and each researcher already solves the resulting best-response problem. Conversely, an equilibrium represented by these Markov stopping policies must satisfy this consistency condition. In the symmetric case,

$$h_{-i}(t) = (n-1)\frac{f(t)}{1 - F(t)}, \tag{16}$$

so one backward problem and one forward law suffice. A finite number of researcher types requires one pair per type, with the appropriate count of opponents of each type. No large-population limit is needed for this reduction.

The baseline has no setup charge: immediate abandonment supplies the entry option. With an additional setup cost K_i, entry compares the initial continuation value with K_i. Mixed entry then requires equality, and nonentry contributes mass at $T_i = \infty$. More generally, randomness of Brownian outcomes is not strategic mixing; a deterministic stopping rule can be optimal. Mixtures must be supported on best responses.

6 Brownian bandits inside a strategic race

The Brownian bandit model [7] uses independent, irreversibly prunable paths, an expected path-time budget, and a terminal-maximum objective. Its population-blind Poisson relaxation couples a backward obstacle problem to a forward survivor calculation through terminal pivotal value. The present game reuses this architecture, replacing the planner's marginal terminal reward by a first-solution prize and replacing the optimization coupling by rival-hazard consistency. Its concavity and optimization certificates do not thereby establish equilibrium convergence.

One researcher can also operate a portfolio of approaches $X_{i\ell}$. Let $A_i(t)$ be the active set, $C_i(A_i)$ its cost rate, and τ_i the first solution among its active approaches. Against rival hazard h_{-i}, the researcher optimizes

$$\sup_{A_i} \mathbb{E}\left[V D_i(0, \tau_i) \mathbf{1}_{\{\tau_i < \infty\}} - \int_0^{\tau_i} D_i(0, t) C_i(A_i(t))\, dt \right], \tag{17}$$

where admissible controls include irreversible pruning and cost becomes zero after total abandonment. A hard compute budget requires a remaining- budget state. This nests an internal allocation problem

inside an external prize race. Correlations among approaches may be modeled internally, but the simple product survival formula still requires independence across researchers.

If compute intensity $u \in [0, \bar{u}]$ time-changes a single approach, $dX = \mu u\, dt + \sigma\sqrt{u}\, dW$ and cost is $cu\, dt$, the continuation operator becomes

$$\partial_t v - (r + h_{-i})v + \sup_{0 \le u \le \bar{u}} u(\mathcal{L}v - c). \tag{18}$$

Here $u = 0$ permits pausing, so this is a separate extension of the baseline. Linearity makes the best intensity an endpoint except at indifference. A finite capacity or increasing marginal compute cost is necessary to avoid an ill-posed instantaneous-acceleration incentive when $\mathcal{L}v - c > 0$.

7 A numerical route and its limits

A finite-deadline approximation suggests the following procedure: initialize rival hazards; solve each backward stopping problem; propagate the induced potential laws forward; recompute hazards using (13); and iterate, with damping if useful. Fixed-point iteration is a proposed solver, not a convergence result. In particular, this is not automatically the concave optimization solved by the Brownian bandit implementation.

Validation should compare the stopping solver with Proposition 3, check conservation of active, abandoned, and solved mass, and measure each researcher's payoff gain from recomputing a best response against the candidate rivals. That unilateral deviation gain, rather than successive-iterate agreement alone, measures approximate Nash behavior. Time, state, and horizon refinement remain necessary.

The main unresolved mathematical tasks are existence and selection of the coupled equilibrium and conditions for convergence of its numerical scheme. Publicly observed rival progress requires a joint-state game; correlated difficulty requires filtering. Neither extension is captured by a deterministic calendar-time hazard alone. The tractable model isolates the central mechanism: the option to keep computing is exercised against an endogenous risk that a competitor makes the prize unavailable.

References

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